Design and Development of a Modular ESP32-Based Rocket Telemetry and Flight Computer System for Experimental Aerospace Applications

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***Abstract*—Rocket telemetry systems are essential for monitoring, control, and post-flight analysis in aerospace and high-power rocketry applications. This paper presents the design and implementation of a modular ESP32-based rocket telemetry and flight computer system capable of real-time telemetry communication, onboard flight-state management, sensor acquisition, autonomous recovery deployment, and SD-card-based flight data logging.**

The proposed avionics platform integrates a BNO055 inertial measurement unit, MS5611 high-resolution altimeter, NEO-M8N GPS module, SX1278 LoRa transceiver, OLED telemetry interface, and dual pyro deployment subsystem into a unified embedded architecture. The firmware employs a modular design methodology to improve scalability, maintainability, and future avionics integration.

Experimental testing demonstrated stable telemetry communication, accurate altitude estimation, reliable orientation sensing, and effective recovery-state detection. The proposed system provides a low-cost yet professional aerospace telemetry solution suitable for educational research, embedded systems experimentation, and amateur rocketry applications.

Keywords— ESP32, Rocket Telemetry, Flight Computer, Aerospace Avionics, LoRa Communication, Embedded Systems, GPS Tracking.

I. Introduction

Modern aerospace systems rely heavily on advanced telemetry and avionics architectures for mission-critical operations and real-time monitoring. Rocket telemetry systems enable continuous acquisition and transmission of critical flight parameters including altitude, orientation, velocity, acceleration, and geographical position.

Commercial aerospace telemetry platforms are often expensive and inaccessible for educational institutions and experimental aerospace projects. Recent advancements in embedded systems, wireless communication technologies, and low-cost sensors have enabled the development of compact and scalable telemetry architectures.

This paper presents the development of a professional ESP32-based rocket telemetry and flight computer system featuring:

- Real-time LoRa telemetry transmission
- GPS-based localization
- High-resolution altitude sensing
- Inertial orientation estimation
- Autonomous recovery deployment
- SD-card flight data logging
- OLED-based flight monitoring

The proposed avionics architecture is intended for experimental aerospace applications, educational research, and advanced embedded systems development.

II. Related Work

Several embedded telemetry systems have been proposed for amateur rocketry and aerospace research. Early systems primarily relied on low-cost microcontrollers such as Arduino Uno and basic RF communication modules.

Recent research has explored the use of ESP32 microcontrollers for aerospace telemetry due to their high computational capability, integrated communication peripherals, and dual-core architecture. LoRa communication technology has also gained popularity for telemetry applications because of its long-range communication capability and low power consumption.

However, many existing low-cost telemetry platforms lack:

- Modular firmware architecture
- Autonomous flight-state management
- Integrated recovery systems
- Expandable avionics infrastructure
- Reliable data redundancy

The RocketTelemetry platform addresses these limitations through a scalable avionics architecture integrating telemetry communication, recovery logic, and onboard data logging.

III. System Architecture

The RocketTelemetry platform is designed around a modular avionics architecture using the ESP32 microcontroller as the primary flight computer.

The system integrates multiple subsystems:

1) Sensor Acquisition Subsystem
2) Telemetry Communication Subsystem
3) Flight-State Management Subsystem
4) Recovery Deployment Subsystem

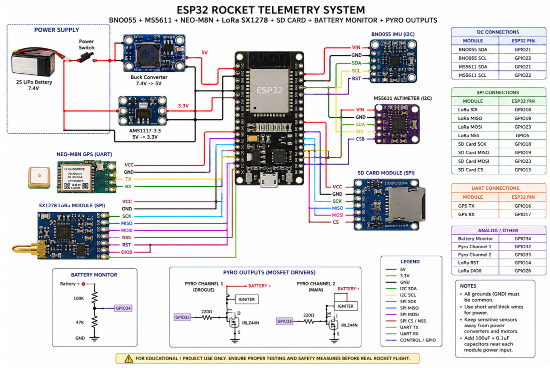

Fig. 1. RocketTelemetry System Architecture

5) Data Logging Subsystem
6) User Interface Subsystem

IV. Hardware Design

A. *ESP32 Flight Computer*

The ESP32 microcontroller serves as the central processing unit of the telemetry system. The dual-core processor enables simultaneous handling of telemetry communication, sensor acquisition, recovery management, and SD-card logging.

B. *Inertial Measurement Unit*

The BNO055 sensor provides orientation estimation including:

- Pitch
- Roll
- Yaw
- Acceleration
- Angular velocity

C. *Altitude Measurement*

The MS5611 high-resolution barometric sensor is used for precise altitude estimation during flight.

D. *GPS Tracking*

The NEO-M8N GPS module provides:

- Latitude tracking
- Longitude tracking
- Satellite count
- Flight path estimation

E. *Telemetry Communication*

The SX1278 LoRa transceiver enables long-range telemetry communication between the rocket and ground station.

V. Firmware Architecture

The firmware was designed using a modular structure to improve scalability and maintainability.

A. *Flight-State Machine*

The telemetry system operates using multiple mission phases:

1) IDLE
2) ARMED
3) BOOST
4) COAST
5) APOGEE
6) DESCENT
7) LANDED

B. *Telemetry Packet Structure*

Telemetry packets contain:

- Altitude
- Velocity
- Orientation
- GPS coordinates
- Battery voltage
- Flight state
- Timestamp

VI. Recovery System

The RocketTelemetry system includes a dual-stage recovery subsystem for autonomous parachute deployment.

The recovery system uses:

- Drogue parachute deployment
- Main parachute deployment
- MOSFET-based switching
- Autonomous apogee detection

VII. Telemetry and Data Logging

Real-time telemetry packets are transmitted using the LoRa communication subsystem while simultaneously logging flight data to the SD card.

TABLE I
TELEMETRY PARAMETERS

Parameter	Description
Altitude	Rocket altitude
Velocity	Vertical speed
Pitch	Orientation angle
Roll	Orientation angle
Yaw	Heading angle
GPS Coordinates	Position tracking
Battery Voltage	System monitoring
Flight State	Mission phase

VIII. Experimental Results

Experimental validation was performed through:

- Sensor acquisition testing
- LoRa communication testing
- Flight-state simulation
- Recovery deployment testing
- SD-card logging validation

The telemetry system demonstrated reliable communication performance and stable sensor acquisition during testing.

IX. Discussion

The developed telemetry platform demonstrates that professional avionics functionality can be achieved using commercially available embedded hardware. The ESP32 microcontroller provided sufficient computational capability for simultaneous telemetry processing and sensor acquisition.

The modular architecture significantly improved firmware maintainability and scalability. LoRa communication enabled stable long-range telemetry transmission with low power consumption.

Although the current implementation achieved stable operation, future improvements may include:

- Kalman filtering
- Adaptive deployment algorithms
- Redundant avionics systems
- CAN-bus integration
- FreeRTOS scheduling
- Web-based telemetry visualization

X. Conclusion

This paper presented the design and implementation of a professional ESP32-based rocket telemetry and flight computer system for experimental aerospace applications.

The proposed system successfully integrates telemetry communication, altitude estimation, orientation sensing, GPS tracking, autonomous recovery deployment, and onboard data logging into a unified avionics platform.

The modular firmware architecture enables scalability and future avionics expansion while maintaining low development cost and high flexibility. The RocketTelemetry platform provides a strong foundation for aerospace education, embedded systems research, and experimental rocketry applications.

Acknowledgment

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